

# Federated Learning for Sleep Stage Classification on Edge Devices via a Model-Agnostic Meta-Learning-Based Pre-Trained Model

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**Abstract**—Sleep stage classification using physiological data obtained from consumer wearable devices has gained significant attention. However, data privacy regulations prevent the collection of individual data for training the models. Federated learning (FL) enables collaborative training across distributed networks of edge devices without sharing private data. This paper presents a federated learning approach that leverages computational resources of Internet of Thing (IoT) edge devices for sleep stage classification. Given the limited computational power and resources on edge devices, we evaluate the impact of model initialization in FL using pre-trained weights of model-agnostic meta-learning (MAML) models. We evaluate the performance of on-device sleep stage classification by comparing local models to different FL settings initialized with random weights (FL-Random) and pre-trained weights (FL-ML). FL-ML enables faster model convergence, yielding better classification performance (up to 22.6%) in a small number of training rounds, compared to local models. Starting from FL-ML reduces the training time required to reach a target accuracy and allows for the training of more accurate models, compared to initializing FL from random weights or using local models. Our study highlights the feasibility of FL initialized with pre-trained weights in real-world IoT settings for sleep stage classification on edge devices.

**Keywords**— Edge device, Federated learning, Meta learning, Sleep stage classification

## I. INTRODUCTION

Sleep is a critical aspect of human life that significantly impacts our daily activities [1], overall health [2], and work productivity [3]. The quality and duration of sleep have profound effects on various physiological activities. Detecting sleep stages, such as nighttime wakefulness (Wake), rapid-eye movement (REM), and non-REM (NREM) sleep, is important for understanding the intricate patterns of human brain activity [4] and metabolism [5] during different phases of sleep. Previous studies have shown sleep detection methods for monitoring sleep quality and stages [6]. At present, polysomnography (PSG) is considered the gold standard technique for assessing sleep quality and states following the technical specifications outlined by the American Academy of Sleep Medicine (AASM) [7]. During a designated sleep time, sleep specialists use PSG to record various physiological signals such as electroencephalogram (EEG), electrocardiogram (ECG), electrooculogram (EOG), and electromyography (EMG). Sleep specialists manually

score these signals to determine the different sleep stages. However, this manual procedure is indeed resource-intensive and time-consuming. Thus, automated methods for sleep stage classification have been developed using machine learning approaches [8], which can significantly reduce the manual effort required.

In recent years, consumer wearable devices have provided the capability to measure physiological signals by wearing them on the body such as on the wrist or fingers [9-11]. These devices have become increasingly popular for monitoring various health and wellness parameters, including heart rate, sleep patterns, physical activity, and more [9, 12]. These physiological signals have served as an alternative resource to the labor-intensive PSG approach [8] for sleep stage classification using machine learning approaches [13, 14]. They provide a more convenient and less burdensome means of gathering sleep-related data [14]. While the cost of classifying sleep stages reduces when using consumer wearable devices for data collection, the limited computational performance and storage capacity of edge devices, such as smartphones and smartwatches, still pose challenges in training models on these devices for widespread adoption. The hardware constraints can impact crucial aspects of model training, including low computation performance and limited storage capacity, thus necessitating careful consideration when developing and deploying such models on edge devices.

Strict data privacy regulations, such as The EU General Data Protection Regulation (GDPR) [15] and the Health Insurance Portability and Accountability Act (HIPAA) [16], present challenges in collecting a sufficient amount of individual data from consumer wearable devices for training machine learning models. To tackle this challenge, we take the first step towards integrating a pre-trained model into federated learning (FL) for sleep stage classification on Internet of Thing (IoT) edge devices. FL is a privacy-preserving machine learning approach that enables training models across multiple decentralized devices while keeping the data localized and private.

Leveraging meta-learning can be highly beneficial for FL on edge devices with heterogeneous training data [17, 18] and constrained resources [19]. We applied a model-agnostic meta-learning (MAML) approach to generate pre-trained weights, which were then used as model initialization in FL

to improve the efficiency and accuracy of sleep stage classification on edge devices while ensuring data privacy and decentralization.

In this work, we aimed to investigate the feasibility and potential of using pre-trained weights to initialize FL for classifying sleep stages in limited recourse settings on edge devices. The main contributions of this paper can be summarized as follows: 1) FL initialized with pre-trained weights from a MAML model across distributed edge devices outperforms the locally trained models without sharing individual data; 2) Initializing FL from pre-trained weights leads to faster convergence, improved performance, and efficient utilization of limited resources in edge computing settings, compared to FL initialized with random weights and the local models.

## II. MATERIALS AND METHODS

### A. Datasets

We utilized raw acceleration data and heart rate from the Apple Watch worn by participants undergoing PSG [13]. The data collection process involved using an app synchronized with the Apple Watch, which recorded motion, heart rate, and a proxy for the circadian clock. A total of 39 subjects were recruited, and they wore the Apple Watch for a duration of 1 to 2 weeks to collect ambulatory steps data. On the final day of the study period, the participants' acceleration and heart rate were recorded during an eight-hour sleep period. PSG signals were sampled at a frequency of 256 Hz and scored according to the standard scoring criteria by AASM. Sleep stages were identified as wake, stage 1, stage 2, and stage 3 of non-REM sleep, and REM sleep based on 30-second epochs while the subjects were in bed. For the classification task in this study, the stage 1, 2, and 3 of non-REM sleep were merged into a single category called NREM. Simultaneously with PSG recording, the raw acceleration and heart rate data were also recorded using the Apple Watch. The acceleration data was measured using a triaxial MEMS accelerometer, which captured acceleration in the x, y, and z directions and was measured in units of  $g$  ( $9.8 \text{ m/s}^2$ ). The heart rate was measured using photoplethysmography, which was recorded on the dorsal aspect of the wrist.

Out of the initial 39 subjects, eight were excluded from the analysis due to transmission errors and the presence of sleep disorders. As a result, the final dataset used in this study consisted of data from 31 subjects. The data included heart rate measurements obtained through photoplethysmography and acceleration data captured by the Apple Watch's triaxial MEMS accelerometer.

### B. Data processing

1) *Heart rate*: The heart rate measurements were initially acquired in beats per minute and sampled at intervals of several seconds. Following the methodology employed by Walch et al. [13], we performed interpolation on the recorded heart rate data, which allowed us to obtain values at a higher frequency, specifically at 1-second intervals. Smoothing and filtering were applied to enhance the heart rate signal and emphasize periods of significant change. Specifically, we convolved the interpolated heart rate data with a difference of Gaussians filter using parameters  $\sigma_1 = 120 \text{ s}$  and  $\sigma_2 = 600 \text{ s}$ . Furthermore, we used a normalization approach to ensure

relative comparability across subjects and mitigate the influence of individual differences in heart rate. We normalized the interpolated heart rate values by dividing them by the 90th percentile of the absolute difference between each heart rate measurement and the mean heart rate over the entire sleep period.

2) *Acceleration*: The acceleration data recorded by the Apple Watch was sampled at a frequency of 50 Hz, except for two cases where the frequency decreased due to power-saving mode or server-side connection issues. To address the missing acceleration data in those two cases, we used interpolation, following the same methodology as described in [13]. Then, we generated activity counts with 1-second intervals, following the approach proposed by Lindert et al. [20].

### C. Sleep Stage Classification with Convolutional Neural Network

We applied a convolutional neural network (CNN) [21] as the model for sleep stage classification. We used 1D convolution layers to capture temporal patterns from the input data. To enhance the model's capability in extracting specific features and reduce computational complexity, we employed grouped convolution, where input channels were divided into multiple groups. This approach promoted interactions within each channel group, enabling the extraction of specialized features while reducing the number of model parameters [22]. Our input data is composed of two channels: heart rate and activity counts. By employing grouped convolution, we captured modality-specific features and enhance computational and memory efficiency. The output of the grouped convolution layer served as the input to the patch embedding process. Patch embedding involved dividing the input into patches and transforming each patch into a vector representation through convolutional operations [23]. This technique reduced the input resolution, thereby lowering computational and memory requirements while efficiently utilizing information from the entire input. Finally, the embedding vectors were passed through three fully-connected layers to classify the sleep stage.

### D. Model-Agnostic Meta-Learning

In this work, we proposed a new initialization strategy for FL based on MAML. MAML is capable of learning both model weights and meta weights, which facilitate the initialization of weights that can be easily adapted to new subject data by considering the potential changes learned from the meta weights [19]. Essentially, the weights obtained from MAML are trained to quickly converge to the global optimum when faced with new tasks or data.

In our work, we used the  $T_1 - T_2$  MAML model [24], which involved the turning of continuous hyperparameters of the model using the gradient of the performance on a validation data called  $T_2$ . For each outer update of the models using the training data, denoted as  $T_2$  the hyperparameters of the L2 penalty, were fine-tuned using stochastic gradient descent (SGD) to minimize the validation cost. This approach ensured that the models were optimized without succumbing to the negative effects of overfitting. After identifying the hyperparameters through random initialization, the model underwent retraining using fixed hyperparameters derived from the final values obtained from  $T_1 - T_2$ .



Fig. 1. An edge computing system using Raspberry Pi 4B (black case) and Raspberry Pi 3B+ (green board)

### E. Federated Learning

We considered two initialization strategies: random initialization (FL-Random) and MAML-based pre-training (FL-ML). The FL-Random model started from randomly initialized parameters that were identical to the parameters of the local model. On the other hand, FL-ML begins with pre-trained parameters obtained from the MAML model. Once initialized, each client in the FL framework started training their respective models using the weights received along with their local data. After completing the local training, the clients transmitted their updated models back to the server. To aggregate the models from all clients, the server employed the FedAvg algorithm [25], which generates an aggregated model by averaging the models' parameters. The aggregated model was then distributed back to each client to serve as the starting point for the subsequent round of training. The iterative training process continued for a predefined number of rounds, as determined by our experimental setup.

## III. EXPERIMENTS

### A. Implementation

We implemented an edge computing training system to simulate real-world conditions. The system comprised five Raspberry Pi 4B and five Raspberry Pi 3B+ as client devices (Fig. 1) and a GPU server as a central server in the FL setup. The specifications of the edge devices used in our experiments are detailed in Table I. To facilitate communication between the edge devices and the server in the FL framework, we employed FedML [26], an open-source library for FL tasks. For the transmission of model weights between the servers and the clients, we used Message Queuing Telemetry Transport (MQTT) protocol [27], which is a widely adopted standard for IoT messaging. MQTT ensured reliable and lightweight messaging, allowing for an efficient exchange of model weights during the FL process.

### B. Training, Validation, and Testing

We evaluated model performance for sleep stage classification using three different models: local, FL-Random, and FL-ML. Among a total of 31 subjects, we selected 10 subjects as a reference dataset. Each subject had unbalanced class labels (Table II). To ensure a fair comparison between the models, we split the dataset of the 10 selected subjects into training and testing sets using an 8:2 ratio. In addition, out of the remaining 21 subjects, 19 subjects were used as the pre-training dataset for inner optimization in the meta-

TABLE I. SPECIFICATION OF THE EDGE DEVICES

|          | Raspberry Pi 3B+                                   | Raspberry Pi 4B                                            |
|----------|----------------------------------------------------|------------------------------------------------------------|
| CPU      | Broadcom BCM2837B0, Cortex-A53 64-bit SoC @ 1.4GHz | Broadcom BCM2711, quad-core Cortex-A72 64-bit SoC @ 1.5GHz |
| RAM      | LPDDR2 1GB                                         | LPDDR4 8GB                                                 |
| GPU      | N/A                                                | N/A                                                        |
| Ethernet | 2.4GHz and 5.0GHz IEEE 802.11 ac wireless          | 2.4 GHz and 5.0 GHz IEEE 802.11ac wireless                 |

learning process. The last two subjects served a dual role, acting as the validation dataset for selecting the best model and the pre-training dataset for outer optimization in the meta-learning process.

We trained all three models using the same configurations with a learning rate of 0.001, weight decay of 0.0001, batch size of 128, and the SGD optimization strategy. For the local model, we trained individual models for each client using only their respective data. During the training process, we recorded the performance at every epoch using a test dataset specific to each client.

For the meta-learning, we initiated an L2 penalty of 0.01 and updated the meta-weights using a learning rate of 0.0001. By using a pre-training dataset comprising 19 subjects, the meta-learning model was updated on specific tasks in the inner optimization process [17]. In the outer optimization process, we optimized the initial parameters of the meta-learning model across subjects. Specifically, we used two subjects for the optimization process. During the meta-learning process, we assessed the performance of the models to select the best-performing model.

For a fair comparison among the three models, we employed one epoch per round in FL training to match the number of updates in local training. In both FL-Random and FL-ML, clients trained their individual models using their respective training datasets. After each round of training, the server aggregated the local models from all clients. We evaluated the performance of the models across clients using the aggregated model at every round. The models were updated until the F1-score of the FL-ML model matched that of the local model, allowing us to compare the effectiveness of improving model performance for the number of updates.

The training process for all models, except the pre-training ML model, was conducted using five Raspberry Pi 4B and five Raspberry Pi 3B+ devices, representing the clients' edge devices. However, for training the ML model, we utilized a single GeForce RTX 3090 GPU server instead of an edge device to handle the computational requirements.

### C. Performance Evaluation Metrics

We evaluated the classification performance of the models on sleep stage classification using the area under the receiver operating characteristic (ROC) curve (AUC) and macro F1-score. We reported the mean and standard deviation of the AUC and macro F1-score values across all clients or devices.

## IV. RESULTS

Fig. 2 shows the classification performance of local and FL models initialized with random weights (FL-Random) and pre-trained weights (FL-ML) on sleep stage classification.

TABLE II. THE NUMER OF CLASSES FOR EACH CLIENT

| Client No. | # of Wake | # of NREM | # of REM |
|------------|-----------|-----------|----------|
| 1          | 2,490     | 10,650    | 3,420    |
| 2          | 510       | 8,820     | 4,830    |
| 3          | 2,460     | 19,890    | 6,990    |
| 4          | 2,760     | 18,150    | 5,970    |
| 5          | 5,340     | 13,740    | 9,270    |
| 6          | 2,760     | 18,660    | 6,240    |
| 7          | 3,090     | 18,720    | 6,750    |
| 8          | 2,490     | 20,730    | 5,400    |
| 9          | 300       | 21,810    | 6,600    |
| 10         | 600       | 20,490    | 7,500    |

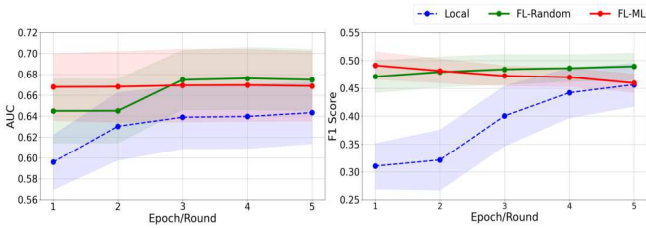


Fig. 2. Performance comparisons of different models on sleep stage classification: local (blue), FL-Random (green), and FL-ML (red). Lines and shaded bands indicate the mean and standard error of mean (SEM) of AUC (left) and F1-scores (right) across 10 edge devices. FL-ML consistently performs better than the local model in both AUC and F1-score throughout the five epochs/rounds. However, FL-Random shows better performance than FL-ML after the third epochs/rounds in terms of AUC and after the second epochs/rounds in terms of F1-score.

Compared to the local models, FL-ML achieved better performance in terms of AUC and F1-score by 12.1% and 58.2% on the first training round and 7.6% and 0.7% on the fifth round, respectively. These findings suggest that FL-ML is more effective in improving model performance during the early stages of training when compared to the local model. FL-Random outperformed FL-ML during the second and third rounds. Specifically, FL-Random improved AUC and F1-score by 6.2% and 1.0%, respectively. These results indicate that FL-Random could provide better performance during some specific training rounds compared to FL-ML. FL-ML performed better than the local model throughout most of the training rounds. However, FL-Random outperformed FL-ML in specific training rounds (3-5 rounds). Overall, FL-ML demonstrates effectiveness in improving model performance during the early stages of training, but there might be specific scenarios or training rounds where FL-Random could provide better performance for sleep stage classification.

Table III shows the training time to train one epoch for the local model and one round for FL-Random and FL-ML. In addition, Fig. 3 shows the classification performance of different models (local, FL-Random, and FL-ML) on sleep stage classification at the first training round. Our results show that FL-ML outperformed both the local and FL-Random models at a fixed training duration for sleep stage classification. It took about 238 s for 7 epochs on Raspberry Pi 4B devices when the local models were trained. FL-ML had

TABLE III. TRAINING TIME FOR EACH MODEL

| Model     | Raspberry Pi 4B   | Raspberry Pi 3B+ |
|-----------|-------------------|------------------|
| Local     | 34 sec (Epoch 1)  | 73 sec (Epoch 1) |
| FL-Random | 199 sec (Round 1) |                  |
| FL-ML     | 224 sec (Round1)  |                  |

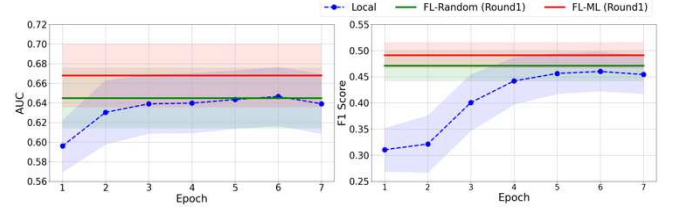


Fig. 3. Performance comparisons of different models on sleep stage classification at the first training round (224 s) : local (blue), FL-Random (green), and FL-ML (red). Lines and shaded bands indicate the mean and standard error of mean (SEM) of AUC (left) and F1-scores (right) across 10 edge devices. The local models took 238 s (epoch 7) on Raspberry Pi 4B and 219 s (epoch 3) on Raspberry Pi 3B+ for training. FL-M achieves the highest performance in terms of AUC and F1-score compared to the local and FL-Random at the first training round.

a higher AUC performance by 3.8% and an improved F1-score of 7.6%, compared to the local model. These results highlight the effectiveness of FL-ML in achieving better performance on Raspberry Pi 4B devices for sleep stage classification. On Raspberry Pi 3B+, the local models were trained only for 3 epochs (about 219 s) due to resource limitations. On this device, FL-ML shows even more improvements in classification performance by 4.5% and 22.6% of AUC and F1-score, respectively, compared to the local model. FL-ML shows superior performance by 3.4% and 8.1% in terms of AUC and F1-score, respectively, compared to FL-Random. These results demonstrate the superiority of FL-ML over FL-Random in achieving better performance on low-resource edge devices.

## V. DISCUSSION AND CONCLUSIONS

In this study, we investigated the impact of initialization strategies in FL for sleep stage classification, considering the limited computational resources of edge devices. We compared locally trained models, FL models with random initialization, and FL models initialized with pre-trained weights from the MAML model. Our findings demonstrated that FL models, despite their decentralized training nature, generally exhibited superior performance compared to locally trained models. Furthermore, we showed that initializing FL models with pre-trained weights from the MAML model yielded faster and more accurate performance in sleep stage classification, compared to both the local and the FL-Random models. Our proposed FL-ML approach showed the best performance, achieving up to a 22.6% improvement within the same training time compared to the other models (local and FL-Random).

The increasing availability of consumer wearable devices presents opportunities for individualized AI-driven services. However, data privacy regulations and limited computational resources of edge devices pose challenges to the widespread utilization of models. In this context, our proposed FL-ML approach, which combines FL with pre-trained MAML weights, may offer a promising solution. By transferring

knowledge from the pre-trained models, the FL-ML approach achieved faster convergence and improved accuracy within the limited computational resources of edge devices.

While our study focused on sleep stage classification, our proposed FL-ML approach can be applied to various tasks relevant to edge devices. By leveraging FL with an efficient model initialization strategy from a pre-trained MAML model, we can effectively overcome the limitations of edge devices and extend the utility of models in resource-constrained environments.

In conclusion, our study presents the significance of initialization strategies in FL and highlights the efficacy of pre-trained models for improved performance in sleep stage classification. Our approach holds promise for addressing data privacy concerns, accommodating limited resources, and enabling a range of AI applications on edge devices.

#### ACKNOWLEDGMENT

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (No. 2021R1C1C1009436), the Institute for Information and Communications Technology Promotion (IITP) grant funded by the Korean government (MSIT) (No. 2017-0-00655), the Korea Health Technology R&D Project through the Korea Health Industry Development Institute (KHIDI), funded by the Ministry of Health and Welfare, Republic of Korea (grant number: HI22C0108), and the Institute for Information and Communications Technology Planning & Evaluation (IITP) grant funded by the Korea government (MSIT) (No. RS-2022-00155911, Artificial Intelligence Convergence Innovation Human Resources Development (Kyung Hee University)).

#### REFERENCES

- [1] G. Medic, M. Wille, and M. E. Hemels, "Short-and long-term health consequences of sleep disruption," *Nature and science of sleep*, pp. 151-161, 2017.
- [2] F. S. Luyster, P. J. Strollo Jr, P. C. Zee, and J. K. Walsh, "Sleep: a health imperative," *Sleep*, vol. 35, no. 6, pp. 727-734, 2012.
- [3] A. J. Krause et al., "The sleep-deprived human brain," *Nature Reviews Neuroscience*, vol. 18, no. 7, pp. 404-418, 2017.
- [4] R. Hastie and B. Park, "The relationship between memory and judgment depends on whether the judgment task is memory-based or on-line," *Psychological review*, vol. 93, no. 3, p. 258, 1986.
- [5] S. Sharma and M. Kavuru, "Sleep and metabolism: an overview," *International journal of endocrinology*, vol. 2010, 2010.
- [6] V. Ibáñez, J. Silva, and O. Cauli, "A survey on sleep assessment methods," *PeerJ*, vol. 6, p. e4849, 2018.
- [7] R. B. Berry, R. Brooks, C. E. Gamaldo, S. M. Harding, C. Marcus, and B. V. Vaughn, "The AASM manual for the scoring of sleep and associated events," *Rules, Terminology and Technical Specifications*, Darien, Illinois, American Academy of Sleep Medicine, vol. 176, p. 2012, 2012.
- [8] I. Perez-Pozuelo et al., "The future of sleep health: a data-driven revolution in sleep science and medicine," *NPJ digital medicine*, vol. 3, no. 1, p. 42, 2020.
- [9] D. J. Miller, C. Sargent, and G. D. Roach, "A validation of six wearable devices for estimating sleep, heart rate and heart rate variability in healthy adults," *Sensors*, vol. 22, no. 16, p. 6317, 2022.
- [10] H. G. van Lier et al., "A standardized validity assessment protocol for physiological signals from wearable technology: Methodological underpinnings and an application to the E4 biosensor," *Behavior research methods*, vol. 52, pp. 607-629, 2020.
- [11] N. Oliver and F. Flores-Mangas, "HealthGear: a real-time wearable system for monitoring and analyzing physiological signals," in *International Workshop on Wearable and Implantable Body Sensor Networks (BSN'06)*, 2006: IEEE, pp. 4 pp.-64.
- [12] S. Saganowski, B. Perz, A. Polak, and P. Kazienko, "Emotion recognition for everyday life using physiological signals from wearables: A systematic literature review," *IEEE Transactions on Affective Computing*, 2022.
- [13] O. Walch, Y. Huang, D. Forger, and C. Goldstein, "Sleep stage prediction with raw acceleration and photoplethysmography heart rate data derived from a consumer wearable device," *Sleep*, vol. 42, no. 12, p. zsz180, 2019.
- [14] S. Djanian, A. Bruun, and T. D. Nielsen, "Sleep classification using Consumer Sleep Technologies and AI: A review of the current landscape," *Sleep Medicine*, 2022.
- [15] P. Voigt and A. Von dem Bussche, "The eu general data protection regulation (gdpr)," *A Practical Guide*, 1st Ed., Cham: Springer International Publishing, vol. 10, no. 3152676, pp. 10-5555, 2017.
- [16] G. J. Annas, "HIPAA regulations: a new era of medical-record privacy?," *New England Journal of Medicine*, vol. 348, p. 1486, 2003.
- [17] C. Finn, P. Abbeel, and S. Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," in *International conference on machine learning*, 2017: PMLR, pp. 1126-1135.
- [18] J. Buckelmüller, H.-P. Landolt, H. Stassen, and P. Achermann, "Trait-like individual differences in the human sleep electroencephalogram," *Neuroscience*, vol. 138, no. 1, pp. 351-356, 2006.
- [19] F. Chen, M. Luo, Z. Dong, Z. Li, and X. He, "Federated meta-learning with fast convergence and efficient communication," *arXiv preprint arXiv:1802.07876*, 2018.
- [20] B. H. Te Lindert and E. J. Van Someren, "Sleep estimates using microelectromechanical systems (MEMS)," *Sleep*, vol. 36, no. 5, pp. 781-789, 2013.
- [21] K. O'Shea and R. Nash, "An introduction to convolutional neural networks," *arXiv preprint arXiv:1511.08458*, 2015.
- [22] Y. Ioannou, D. Robertson, R. Cipolla, and A. Criminisi, "Deep roots: Improving cnn efficiency with hierarchical filter groups," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 1231-1240.
- [23] A. Dosovitskiy et al., "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale," 2020-10-22T17:55:59 2020.
- [24] J. Luketina, M. Berglund, K. Greff, and T. Raiko, "Scalable gradient-based tuning of continuous regularization hyperparameters," in *International conference on machine learning*, 2016: PMLR, pp. 2952-2960.
- [25] C. Wu, F. Wu, L. Lyu, Y. Huang, and X. Xie, "Communication-efficient federated learning via knowledge distillation," *Nature Communications*, vol. 13, no. 1, 2022-04-19 2022, doi: 10.1038/s41467-022-29763-x.
- [26] C. He et al., "Fedml: A research library and benchmark for federated machine learning," *arXiv preprint arXiv:2007.13518*, 2020.
- [27] U. Hunkeler, H. L. Truong, and A. Stanford-Clark, "MQTT-S—A publish/subscribe protocol for Wireless Sensor Networks," in *2008 3rd International Conference on Communication Systems Software and Middleware and Workshops (COMSWARE'08)*, 2008: IEEE, pp. 791-798.